

Tutorial Questions (Set-II)

Q1 A set X is said to be *infinite* if there is an injective map $f: X \rightarrow X$ which is not surjective. A set X is said to be *finite*, if it is not infinite. A set X is said to be *denumerable* (or *countably infinite*) if there is a bijection $f: \mathbb{N} \rightarrow X$. By a *countable* set, we mean a set which is either finite or denumerable.

- (i) Let X be an infinite set and $x \in X$. Show that $X \setminus \{x\}$ is also infinite. Further, show that X contains a denumerable subset.
- (ii) Show that $\mathbb{N} \times \mathbb{N}$ is denumerable. Deduce that if A and B are countable, then cartesian product $A \times B$ is also countable. Hence or otherwise, show that the set $\mathbb{Q}$ of rational numbers is countable.
- (iii) Show that $[0, 1]$, $(0, 1)$ and $\mathcal{P}(\mathbb{N})$ are uncountable.

Q2 Consider the n -dimensional vector space $\mathbb{R}^n$. For $a = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$, define

$$\|a\|_1 = \sum_{i=1}^n |a_i|, \quad \|a\|_2 = \left(\sum_{i=1}^n a_i^2 \right)^{\frac{1}{2}} \quad \text{and} \quad \|a\|_\infty = \max\{|a_i| : 1 \leq i \leq n\}.$$

- (i) For $a = (a_1, a_2, \dots, a_n), b = (b_1, b_2, \dots, b_n) \in \mathbb{R}^n$, prove Lagrange's identity

$$\left(\sum_{i=1}^n a_i b_i \right)^2 = \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right) - \sum_{1 \leq i < j \leq n} (a_i b_j - a_j b_i)^2.$$

Hence, deduce Cauchy-Schwarz inequality $|\langle a, b \rangle| \leq \|a\|_2 \|b\|_2$.

- (ii) Show that $\|\cdot\|_1, \|\cdot\|_2$ and $\|\cdot\|_\infty$ are norms on $\mathbb{R}^n$. Further, show that these are equivalent norms on $\mathbb{R}^n$. Hence, deduce that the set of open sets in all the associated metric spaces $\mathbb{R}^n$ are the same.
- (iii) Let $p \in \mathbb{R}$ with $1 \leq p < \infty$. Set $q \in \mathbb{R}$ with $\frac{1}{p} + \frac{1}{q} = 1$. If $p = 1$, then $q = \infty$. (q is called *conjugate exponent* of p .) For $\alpha, \beta \in \mathbb{R}^+$ and $p > 1$, show that $\alpha\beta \leq \frac{\alpha^p}{p} + \frac{\beta^q}{q}$. Hence, prove the Hölder's inequality

$$\sum_{i=1}^n |a_i b_i| \leq \left(\sum_{i=1}^n |a_i|^p \right)^{\frac{1}{p}} \left(\sum_{i=1}^n |b_i|^q \right)^{\frac{1}{q}}.$$

- (iv) Show that $\|a\|_p = \left(\sum_{i=1}^n |a_i|^p \right)^{\frac{1}{p}}$ defines a norm on $\mathbb{R}^n$ called the ℓ^p -norm. Also, show that $\|a\|_\infty = \lim_{p \rightarrow \infty} \|a\|_p$.

Q3 For $x, y \in \mathbb{R}$, let d_1 and d_2 be defined as

$$d_1(x, y) = \sqrt{|x - y|} \text{ and } d_2(x, y) = \begin{cases} |x - y| + 1, & \text{if } x > 0, y \leq 0 \text{ or } y > 0, x \leq 0, \\ |x - y|, & \text{otherwise.} \end{cases}$$

- (i) Show that d_1 and d_2 are metrics on $\mathbb{R}$.
- (ii) Let (X, d) be a metric space. Show that ρ and η given by

$$\rho(x, y) = \frac{d(x, y)}{1 + d(x, y)} \quad \text{and} \quad \eta(x, y) = \max\{d(x, y), 1\}, \quad \text{for } x, y \in X$$

are metrics on X . Further, show that d, ρ and η are equivalent metrics on X .

Q4 Let (X, d) be a metric space and $A \subseteq X$. Let $D(A)$ be the set of accumulation points of A (called the *derived set* of A).

- (i) Show that $a \in D(A)$ if and only if there exists a sequence $(a_n)_{n \in \mathbb{N}}$ of distinct elements of A with $\lim_{n \rightarrow \infty} a_n = a$.
- (ii) If $A \subseteq B \subseteq X$, then show that $D(A) \subseteq D(B)$. Further, show that $D(D(A)) \subseteq D(A)$, i.e., $D(A)$ is closed subset of X .
- (iii) For $A, B \subseteq X$, show that $D(A \cup B) = D(A) \cup D(B)$ and $\overline{(A \cup B)} = \overline{A} \cup \overline{B}$, where $\overline{A} = A \cup D(A)$ is the *closure* of A in X .
- (iv) For $A \subseteq X$ and $x \in X$, define $d(x, A) = \inf\{d(x, a) : a \in A\}$. Show that $\overline{A} = \{x \in X : d(x, A) = 0\}$.

Q5 Let $f: (X, d) \rightarrow (Y, \rho)$ be a function. Then show that f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq X$.

- Q6
- (i) Show that $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$; $a < b$ is homeomorphic with $[0, 1]$.
 - (ii) Show that any open interval $(a, b) = \{x \in \mathbb{R} : a < x < b\}$; $a < b$ or $(-\infty, b)$ or (a, ∞) or $\mathbb{R} = (-\infty, \infty)$ are all homeomorphic with $(0, 1)$.
 - (iii) Show that closed interval $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$ and open interval $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ for $a < b$ are not homeomorphic.

Q7 Show that the n -dimensional Euclidean space $\mathbb{R}^n$ is a complete metric space.